Machine Learning

Bachelor in Applied Mathematics and Computing

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Data Mining Methodology

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In this Session:

- Data Selection and Transformation
- Hypothesis Evaluation
- Hypothesis Contrast: t-test

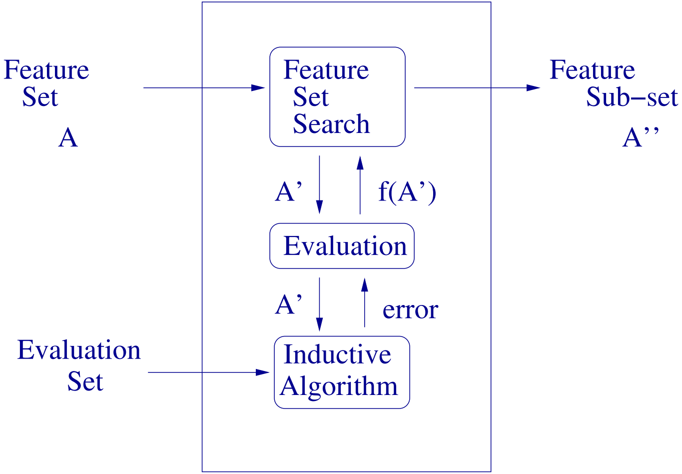
Selection of input data and pre-process

- Data Selection
  - Random
  - The most similar
  - The most different
  - Folowing a distribution
  - In class borders
  - Weighting data proportionally to its classification error: *Boosting*
  - Incremental: incorporate data continuously
- Pre-processing
  - Noise reduction
  - Unknown data threatment

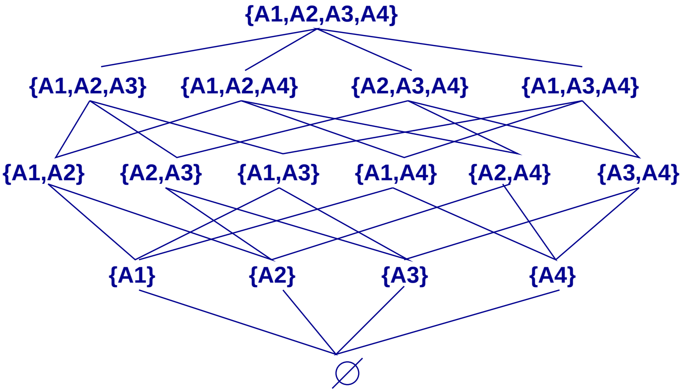
Feature Selection

- Dimensionality reduction
- Techniques:
  - Feature Ranking/Filtering: order the features individually following some criteria
  - Sub-set selection: search in the space of feature sets
  - Feature selection and model learning in one step
- Search: any search algorithm
- Evaluation: correlation, entropy, mutual information, ...
- Stop Criteria: percentage, threshold, iterations, ...

Feature Selection with Wrappers



Wrapper Example



Wrapper Building as Search

- Search in the space of all the possible feature sets
- Begin with
  - complete feature set
  - empty feature set
- Search types
  - Hill-climbing search
  - Best-first search
- The evaluation of each node (feature sub-set) is performed calling to the selected inductive learning algorithm

Feature Selection: PCA

- PCA: Principal Component Analysis
- For numeric attributes
- Consists on a projection of the original feature space in a reduced one: $x_1, x_2, \ldots, x_n \rightarrow f_1, f_2, \ldots, f_p$, where $p \leq n$
- Characteristics of each f_i:
  - Mutual independence
  - Ordered by relevance (information content)
- The ordering permits to select a sub-set k of relevant features
- We loose the semantic of the features

Feature Selection

- Pearson correlation coefficient:

$$r_{x,y} = \frac{\sum(x_i - \mu_x)(y_i - \mu_y)}{(n-1)\sigma_x\sigma_y}$$

  where n is the number of examples, μ_i is the average of feature i and σ_i is the standard deviation
- Mutual Information
  - Discrete case:

$$I_{x,y} = \sum_y \sum_x p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

  - Continuos case:

$$I_{x,y} = \int_y \int_x p(x,y) \log \frac{p(x,y)}{p(x)p(y)} dxdy$$

Feature Generation

- Dynamic generation of significant attributes:
  - Comparison of attributes with same type
  - Discretization of continuous features in intervals
  - Cluster several nominal values in set of values
  - Count the number of features that are true and/or false in a set of boolean features
  - Cluster two attributes that appear in one rule
  - Cluster features or feature values that appears in a rule in such a way that exactly cover the possitive examples of the rule
  - Feature Combinations:
    - Continuous features: mathmatical operations (sum, substraction, multiplication, average, . . .)
    - Discrete attributes: logic operations (conjunction, disjunction, . . .)

More Pre-processing Operations

- Discretization: unsupervised learning methods over one or several features
- Numerization: "1 a n": One binary feature for each original nominal value of a feature
- Range normalization: for instance, lineal ($v' = \frac{v - min}{max - min}$)
- Other dimensionality reduction techniques like self-organizing maps

Introduction

- Hypothesis evaluation: estimate the quality of the learned model
- Motivation:
  - Decide whether to use the model or not:
    Should we consider as good a model generated with 40 training instances?
  - Hypothesis evaluation is a main issue in many learning algorithms:
    Is adequate to prune a decision learned tree?

Challenges in hypothesis evaluation

- Bias in the estimation: estimations performed over the whole training set may be poor when trying to predict the success in future data (overffiting to training set)
- Variance in the estimation:
  - Estimations performed over test sets also generate differences over real future results
  - Different test sets may produce different results

Hypothesis Estimation

- Given:
  - a hypothesis/model h,
  - a set of data with n examples generated randomly following a distribution $\mathcal{D}$,
- What is the best estimation of the success of h over future instances that also follow distribution $\mathcal{D}$?
- What is the most probable error in such estimation?

Error types

- *Error in examples (or sample)* of a hypothesis h with respect to a goal function f and a set of data S:

$$error_S(h) = \frac{1}{n} \sum_{x \in S} \delta(f(x), h(x)) \tag{1}$$

  where n is the size of the set S, y $\delta(f(x), h(x)) = 0$ if $h(x) = f(x)$, and 1 otherwise

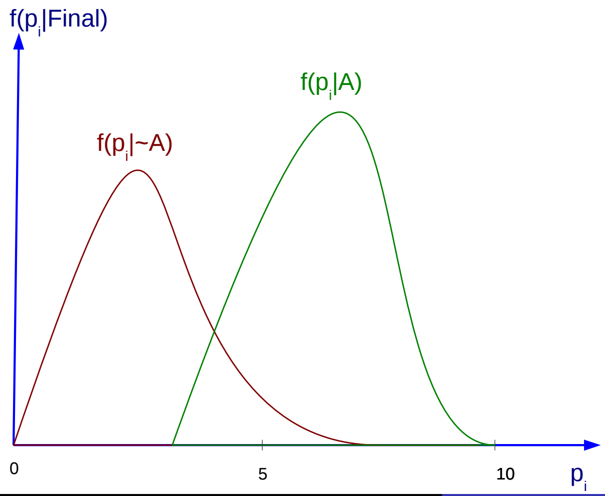
- *Real Error* of a hypothesis h with respect to a goal function f and a distribution $\mathcal{D}$ is the probability that h does not classify correctly an instance that follows distribution $\mathcal{D}$:

$$error_D(h) = Prob_{x \in \mathcal{D}}[f(x) \neq h(x)] \tag{2}$$

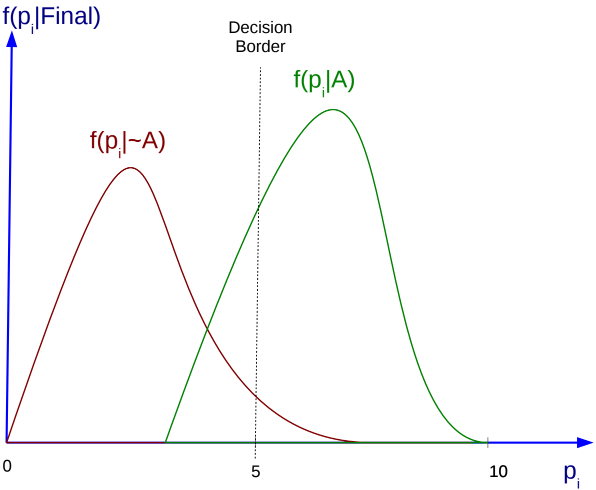
Errors in Samples

- Given:
  - Last Year Course:
    - Marks of the students in class exercises: $p_i \in [0..10]$
    - Marks of the students in final exam: $t_i \in [0..10]$
    - Final Marks: $if((p_i + t_i)/2 \geq 5; A; \sim A)$
  - Current Course:
    - Qualifications of the students in class exercises: $p_i \in [0..10]$
- Query: can I predict the final qualification without correcting the final exam?

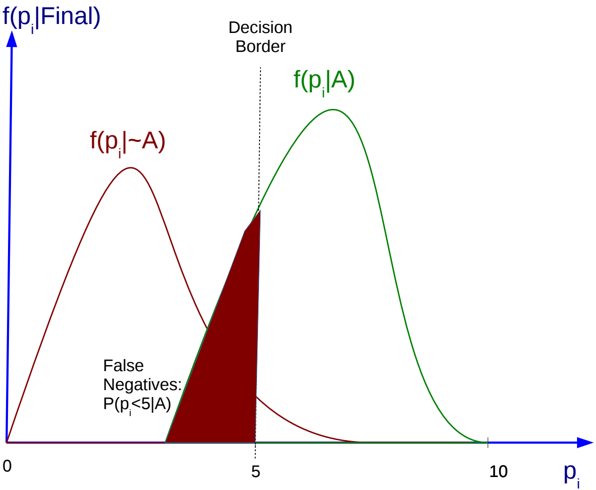
Conditional Probability Density Functions



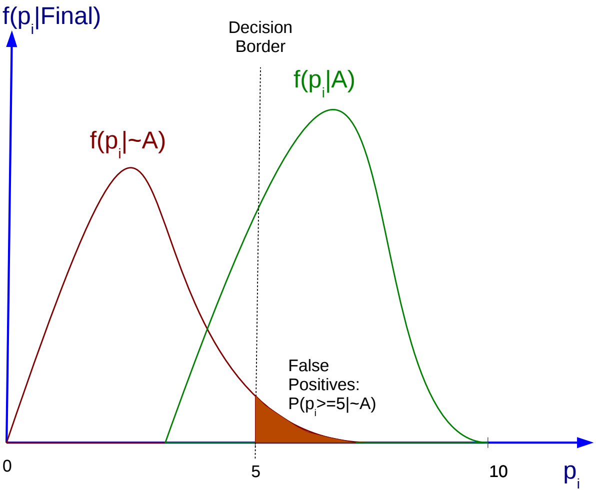
Decision Border



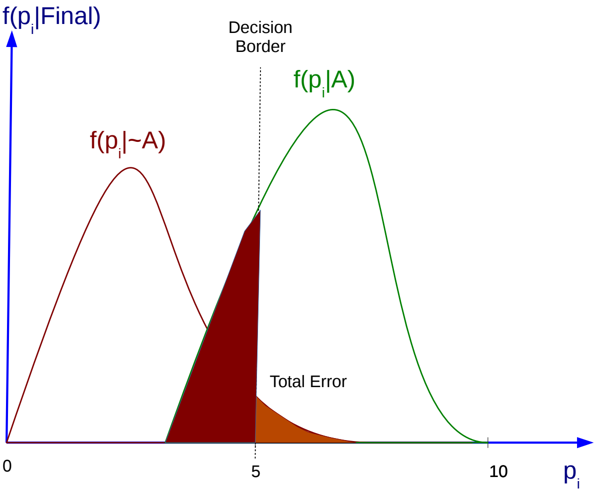
False Negatives



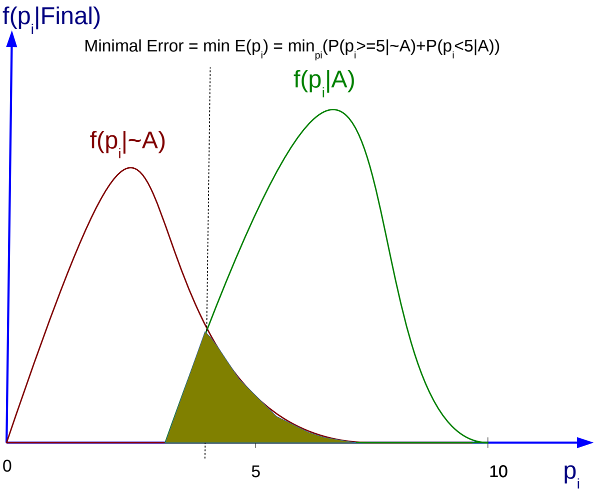
False Positives



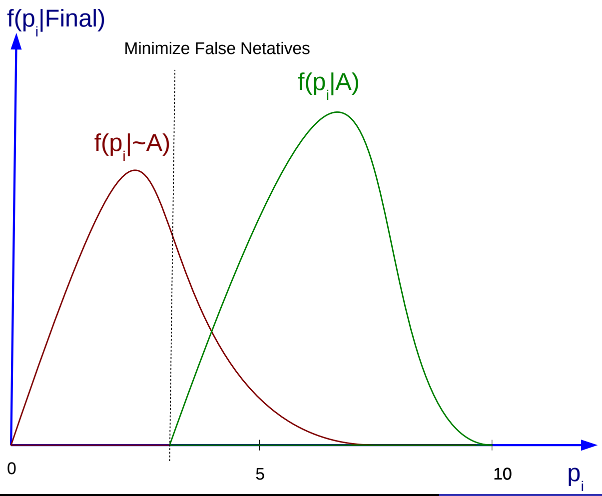
Total Error



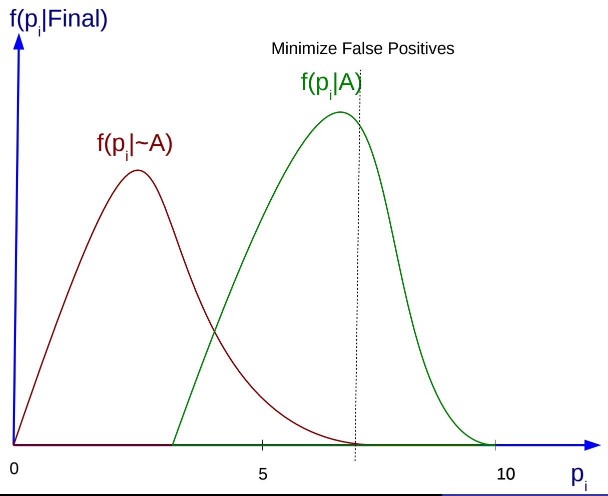
Minimum Error



Minimize False Netatives

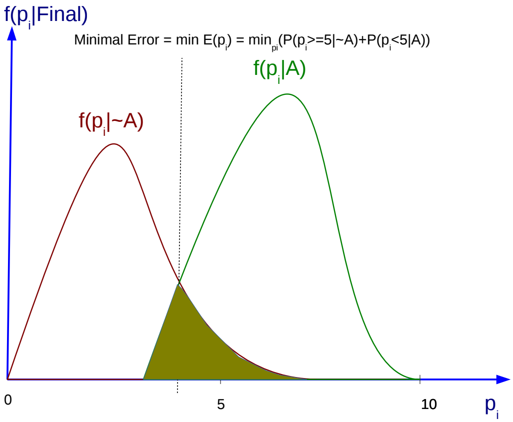


Minimize False Positives



Real Error

- Guess we choose next hypothesis:



Confusion Matrix

		Actual class	
		positive	negative
Predicted class	positive	Real Positive (TP)	False Positive (FP)
	negative	False Negative (FN)	True Negative (TN)

Metrics (1)

- Global Precision (*accuracy*): total correct answers

$$accuracy = \frac{TP + TN}{P + N}$$

Metrics (2)

- Precision of the class (*precision*):

$$precision = \frac{TP}{TP + FP}$$

How many of the predicted ones are correct?

- Sensitivity (*recall*): Rate of true positives

$$recall = \frac{TP}{TP + FN} = \frac{TP}{P}$$

How many of the ones that I should predict, are actualy predicted?

Metrics (2)

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$$precision = \frac{TP}{TP + FP}$$

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How many of the predicted ones are correct?

How many of the ones that I should predict, are actualy predicted?

Metrics (3)

- Specificity: Rate of true negatives

$$specificity = \frac{TN}{TN + FP} = \frac{TN}{N}$$

- Rate of false positives (1 - specificity)

$$1 - specificity = \frac{FP}{FP + TN} = \frac{FP}{N}$$

How many of the excluded ones are actually correct?

How many of the ones I should exclude are being chosen by mistake?

Metrics (3)

- Specificity: Rate of true negatives

$$specificity = \frac{TN}{TN + FP} = \frac{TN}{N}$$

- Rate of false positives (1 - specificity)

$$1 - specificity = \frac{FP}{FP + TN} = \frac{FP}{N}$$

How many of the excluded ones are actually correct?

How many of the ones I should exclude are being chosen by mistake?

Metrics (4)

- F-Measure

$$F - Measure = 2 \times \frac{precision \times recall}{precision + recall}$$

- kappa coefficient
- Matthews correlation coefficient

Cross Validation: Confidence Interval $error_D(h)$

- A confidence interval of a $N\%$ of a parameter p is an interval expected to contain p with a probability of $N\%$
- Computing the confidence interval of $error_D(h)$:
  - $error_S(h)$ follows a binomial distribution
  - The average of the distribution is $error_D(h)$
  - The standard deviation is $\sqrt{\frac{error_S(h)(1-error_S(h))}{n}}$
  - Find the interval centered in the average which is large enough to contain the $N\%$ of the probability of the distribution.
- Complex computation for a binomial distribution
- But easy for a normal one, which approximates very accurately the binomial distribution of same average and standard deviation when n is large enough

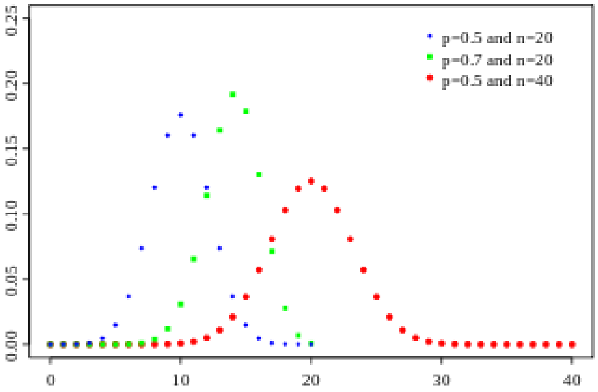
Binomial Distribution

- Provides the probability to observe r heads in a population of n independent flips of a coin, when the probability to get a head is p.
- In other words: provides the probability to get r fails when a test over n examples is performed and when the real probability of fail is p.

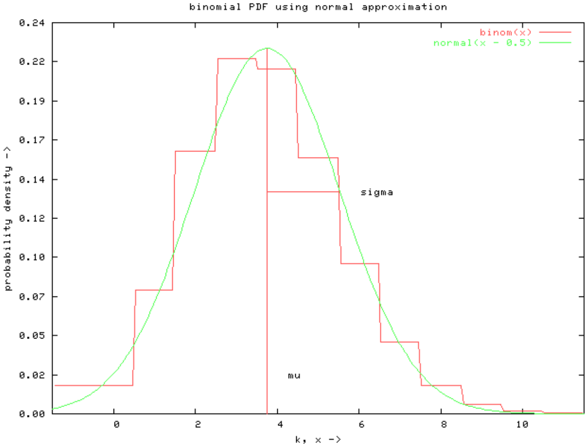
$$P(r) = \frac{n!}{r!(n-r)!} p^r (1-p)^{n-r} \tag{3}$$

- Expectation: $E[X] = np$
- Variance: $Var(X) = np(1-p)$
- Standard deviation: $\sigma_X = \sqrt{np(1-p)}$

Examples of Binomial Distributions

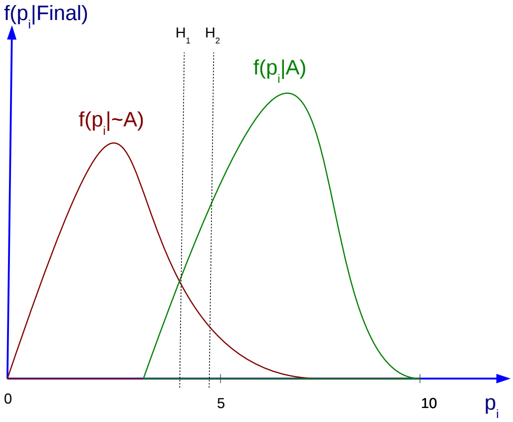


Binomial ($n = 25, p = 0.15$)



Hypothesis Contrast

- Guess we have two different hypothesis (learned models):



Cross Validation for t-test

1. Split the data set, D_0 in K disjunt sets, $T_1, T_2, \ldots, T_k$ with equal size, and where this size must be larger than 30
2. For $i = 1$ to K do
   - Use T_i as test set, and the rest of data for training, S_i:
     - $S_i \leftarrow \{D_0 - T_i\}$
     - $h_A \leftarrow L_A(S_i)$
     - $h_B \leftarrow L_B(S_i)$
     - $\delta_i = error_{T_i}(h_A) - error_{T_i}(h_B)$
3. Return $\bar{\delta}$ where:

$$\bar{\delta} \equiv \frac{1}{k} \sum_{i=1}^{k} \delta_i$$

Paired t-test

- Goal:
  - Estimate $E_{S \in D_0}[error_{\mathcal{D}}(L_A(S)) - error_{\mathcal{D}}(L_B(S))]$
  - i.e. analize whether the differences between $error_{\mathcal{D}}(L_A(S))$ and $error_{\mathcal{D}}(L_B(S))$ are significant from the statistical point of view
- With a confidence interval of $N\%$, the previous value is in the range:

$$\bar{\delta} \pm t_{N,k-1} S_{\bar{\delta}}$$

Where:

- $t_{N,k-1}$ is a constant that plays a similar role thanz_N in a t distribution
- $S_{\bar{\delta}}$ is an estimation of the standard deviation of the distribution t that governs $\bar{\delta}$:

$$S_{\bar{\delta}} \equiv \sqrt{\frac{1}{k(k-1)} \sum_{i=1}^{k} (\delta_i - \bar{\delta})^2}$$